

# GURUVAIYA MUTHUKARUPPAN

## Principal SDET | Test Automation Architect

Chennai, India | +91 6381034340 | guru4ec@gmail.com | linkedin.com/in/guruvaiya-m | github.com/guru4ec-dev | Open to Remote  
75% Regression Reduction • 40% Fewer UAT Defects • 97% Grid Stability • Enterprise-Scale Automation

### PROFESSIONAL SUMMARY

Principal SDET and Test Automation Architect with 19+ years building enterprise-scale automation frameworks for Fortune 500 platforms across insurance, banking, retail, and telecom. Hands-on expertise in Java, Selenium, Playwright, REST Assured, Selenium Grid 4, Docker, Jenkins, and GitHub Actions. Proven track record architecting frameworks from scratch, cutting regression execution time by 75%, and driving AI-assisted quality engineering practices with GitHub Copilot and Claude Code.

### CAREER HIGHLIGHTS

- 19+ years of enterprise test automation engineering across Fortune 500 organizations, including Home Depot, JPMorgan Chase, Bank of America, and Allianz.
- 9+ years working onsite in the USA and Australia, delivering automation directly alongside client engineering and business teams at Home Depot, Commonwealth Bank, and Vodafone.
- Reduced regression execution time by 75% (4 hours → 1 hour) by re-architecting parallel test execution.
- Improved Selenium Grid execution stability from 60% to 97% through Docker-based Selenium Grid 4 architecture.
- Built and maintained 500+ automated UI and API regression scenarios across four enterprise applications.
- Automated 60+ REST APIs with REST Assured, embedding validation into CI/CD quality gates.
- Cut integration defects reaching UAT by 40% through shift-left API contract testing.
- Enabled automation adoption across 8 Agile squads and mentored 5+ QA engineers.
- Improved engineering productivity by 30–40% through AI-assisted development with GitHub Copilot and Claude Code.

### CORE TECHNICAL SKILLS

**Automation Architecture:** Framework Design, Selenium WebDriver, Java, TestNG, Cucumber BDD, Page Object Model, Hybrid Frameworks

**UI/API Automation:** REST Assured, Playwright (Java, TypeScript — in progress), API Contract Testing, Postman/Newman, Swagger/OpenAPI, SQL, JSON/XML

**CI/CD & DevOps:** Jenkins, GitHub Actions, Docker, Selenium Grid 4, Maven

**AI-Assisted Engineering:** GitHub Copilot, Claude Code, Enterprise Gen AI, AI-Assisted Test Authoring

**Test Strategy & Design:** Data-Driven Testing, Shift-Left Testing, Parallel Execution, Cross-Browser Testing

**Tools & Practices:** JIRA, Xray, Zephyr, Git, Agile/Scrum, ROI Analysis, Defect Management, Mentoring

### PROFESSIONAL EXPERIENCE

#### Principal SDET & Automation Architect — Allianz (Enterprise Web & API Automation)

Tata Consultancy Services • Jan 2024 – Present • Chennai, India

- Architected a reusable automation framework from scratch across four Allianz France platforms — Espace Client Entreprise, ECAR (Retirement Portal), ECP (Professional Customer Portal), and Customer Identity & Access Management (CIAM).
- Built and maintained 500+ automated UI and API regression scenarios across the four applications, increasing regression coverage and release confidence.
- Re-architected parallel test execution using Selenium Grid 4 + Docker, cutting regression runtime from 4 hours to 1 hour (75%) and raising the share of test runs completing without Grid/infrastructure failures from 60% to 97%.
- Automated 60+ REST APIs with REST Assured, embedding validation into Jenkins and GitHub Actions quality gates enforced before every merge — cutting integration defects reaching UAT by 40%.
- Introduced Playwright (Java) for new test development alongside the existing Selenium suite, using GitHub Copilot, Claude Code, and enterprise Gen AI tooling for AI-assisted script authoring and coverage-gap analysis.
- Enabled framework adoption across 8 Agile squads, serving as framework owner and technical advisor for 4 dedicated squads.
- Improved regression efficiency by increasing automation coverage and reducing dependency on manual validation cycles.
- Drove a measured 30–40% improvement in engineering productivity by adopting GitHub Copilot and Claude Code for test generation, refactoring, and failure analysis.

- Mentored 5 QA engineers and standardized automation practices across 8 cross-functional teams, cutting onboarding time by 33% (12→8 weeks).

## **Test Automation Architect | Automation Test Manager — Home Depot (Fortune 500 E-Commerce & Supply Chain)**

*Tata Consultancy Services · Jan 2019 – Nov 2023 · Atlanta, GA, USA*

- Owned end-to-end automation architecture for one of the US's highest-traffic retail platforms, safeguarding checkout, cart, and inventory journeys with zero critical production defects.
- Designed an organization-wide automation strategy and CI/CD quality gate framework adopted across all concurrent product streams, accelerating release cadence company-wide.
- Directed a team of 4–6 SDETs through structured code reviews and best-practice coaching, enabling multiple major product launches without quality regressions.

## **Automation Test Lead — Commonwealth Bank (CBA) — Financial Crime, Future Advice & Customer Engagement**

*Tata Consultancy Services · Nov 2015 – Nov 2018 · Sydney, Australia*

- Delivered automation for CBA's Financial Crime & Sanctions platform (Fircosoft/SWIFT/NPP) with zero compliance-related production incidents, protecting Australia's largest bank from regulatory exposure.
- Built a full BDD automation suite for the Future Advice Model (FAM), scaling regression coverage for a platform serving thousands of financial advisers.
- Surfaced 15+ data-sync defects invisible to manual testing by integrating SQL backend validation into the Customer Engagement Platform automation suite.

## **Automation Test Lead — Vodafone Australia — ATG/AWS E-Commerce Platform**

*Tata Consultancy Services · Jan 2015 – Oct 2015 · Sydney, Australia*

- Built a Selenium WebDriver + Java automation suite from scratch, unblocking fortnightly releases that had stalled on manual regression bottlenecks.
- Prevented 15+ scope deviations across purchase, plan upgrade, and account management journeys as technical bridge between development, BA, and QA teams.

## **Senior Automation Engineer — TCS Automation Centre of Excellence — Framework Development & Strategy**

*Tata Consultancy Services · May 2011 – Dec 2014 · Chennai, India*

- Co-developed the RAFT, TAM, and eRAFT automation frameworks, adopted globally by JPMorgan Chase, Bank of America, Nielsen, Nationwide, and Humana across 6 industry verticals.
- Led 5–6 proof-of-concept engagements per year for Fortune 500 accounts, producing ROI analyses and automation roadmaps that directly influenced client adoption decisions.
- Trained 20+ engineers across TCS accounts through structured in-house programs, accelerating automation capability ramp-up and reducing external vendor dependency.

## **Automation QA Engineer — Softeon India Pvt Ltd — SCflex Enterprise Supply Chain Management (SONY, Ben E. Keith, Network Global Logistics)**

*2006 – 2011 · Chennai, India*

- Built end-to-end QA coverage from scratch across 5 core enterprise modules (Warehouse, Transportation, Yard, Order, and Parcel Management) for US clients including SONY.
- Designed and executed cross-browser and cross-platform test suites, safeguarding consistent behavior across a J2EE, database-independent platform serving multiple US clients.

## **CERTIFICATIONS**

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Google Cloud Platform (GCP) — Certified

## **EDUCATION**

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B.Tech — Electronics Engineering | MIT, Anna University, Chennai

## **RECOGNITION**

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- Client appreciation awards for automation delivery at Allianz and Home Depot engagements.
- Multiple "On the Spot" awards at Tata Consultancy Services for delivery excellence.
- Recognized for resolving critical production issues across global engagements.